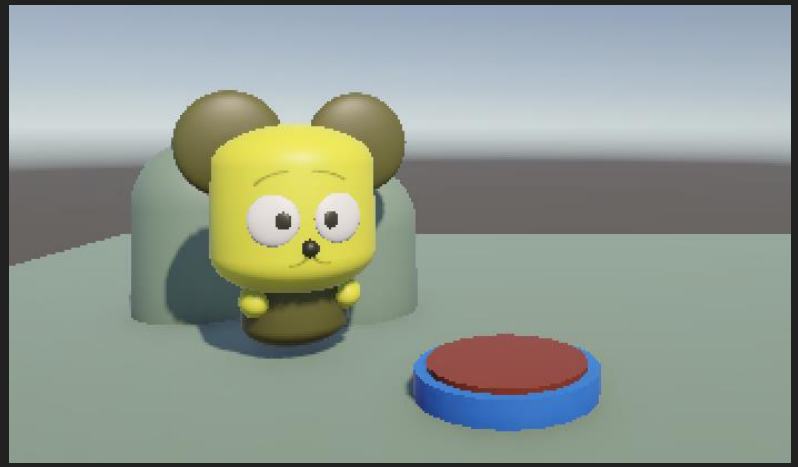
Game systems

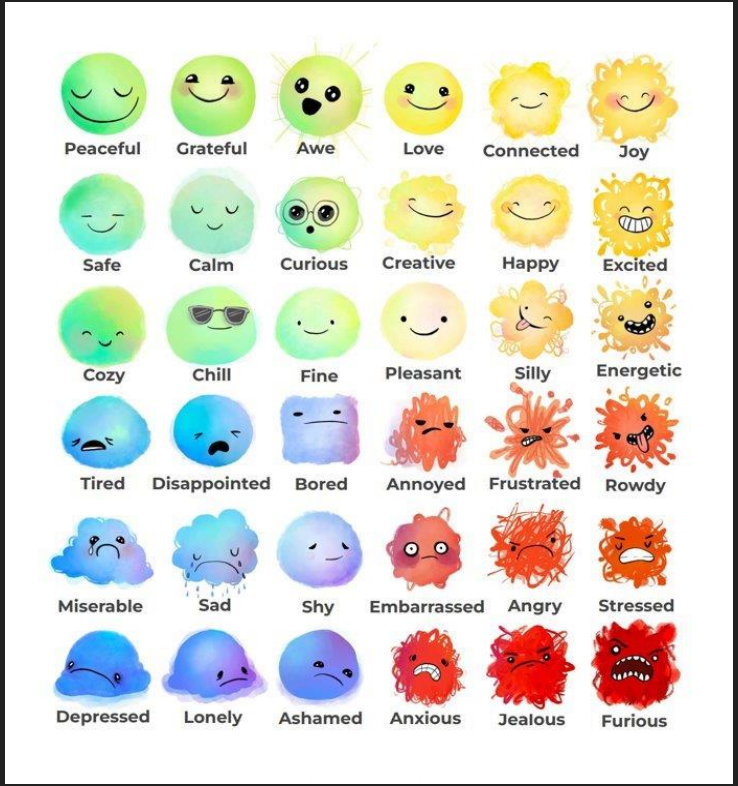
Virtual Pet (part 2)



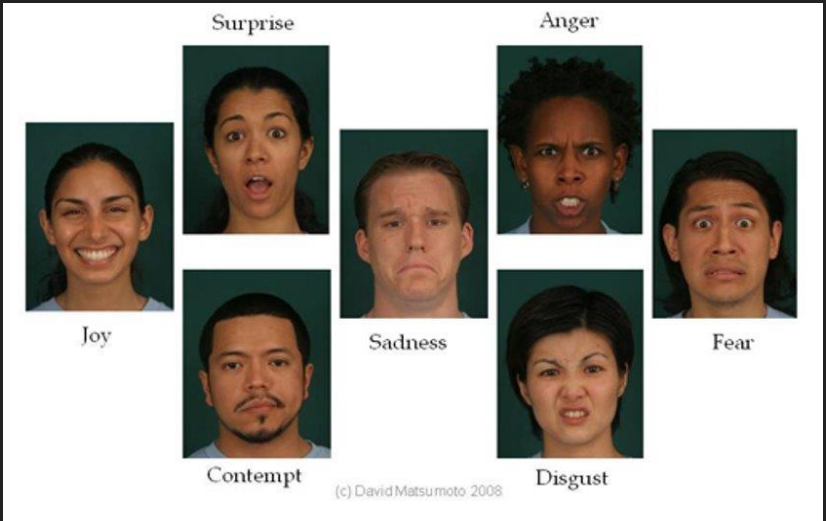
Result so far



Non-Verbal Communication: A Designer's Tool



Facial Expressions: A Window to the Soul



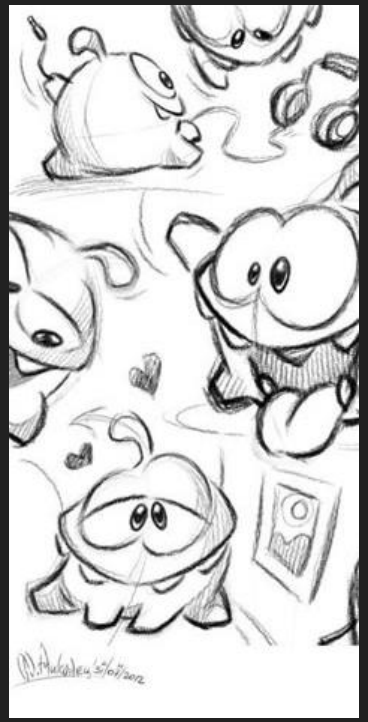
- Eyes: Are they wide with joy, or narrowed with suspicion?
- Mouth: Is it a soft smile of contentment, or a frown of disappointment?
- Brows: Are they furrowed in anger, or relaxed in a state of calm?

Ekman, P. (1992). Facial Expressions of Emotion.

Checklist for Non-Verbal Pet Communication

Face:

- Eyes: The shape and size of the eyes can instantly convey emotion. Consider wide, open eyes for surprise or narrowed eyes for happiness or suspicion.
- Eyebrows: Eyebrows are extremely expressive. Use them to furrow (anger, confusion), raise (surprise), or relax (calmness). If your pet doesn't have eyebrows, a facial marking or feature that can be animated will achieve a similar effect.
- Mouth: A mouth that can change shape is essential for showing basic emotions, from a happy smile to a disappointed frown.



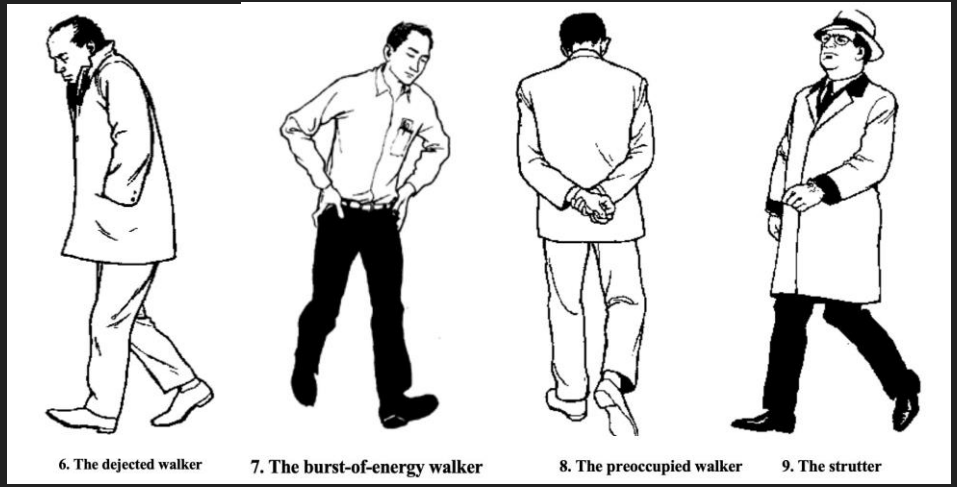
Checklist for Non-Verbal Pet Communication

Body:

- Back/Spine: The ability to arch or straighten the back is fundamental for posture. A curved back can show sadness or fear, while a straight, proud back conveys confidence.
- Neck: The pet's head should be able to move independently of its body. A lowered head suggests sadness, while a tilted head shows curiosity.
- Limbs: Limbs should be able to move towards or away from the body. Tucked-in limbs can show shyness or defensiveness, while wide-open limbs communicate openness and joy.
- Tail: If your pet has a tail, it is a powerful expressive tool. It can wag with excitement, go stiff with alertness, or be tucked between the legs with fear.

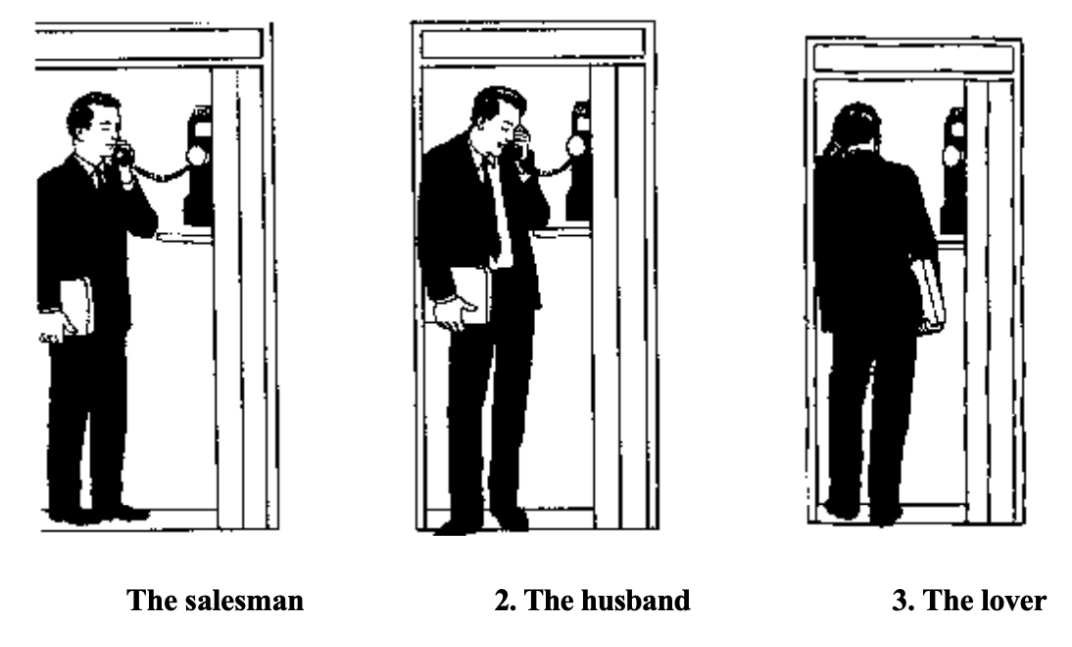


Body Language: Posture and Personality



https://www.youtube.com/watch?v=QJkAEfmBnZM&t=191s

Body language

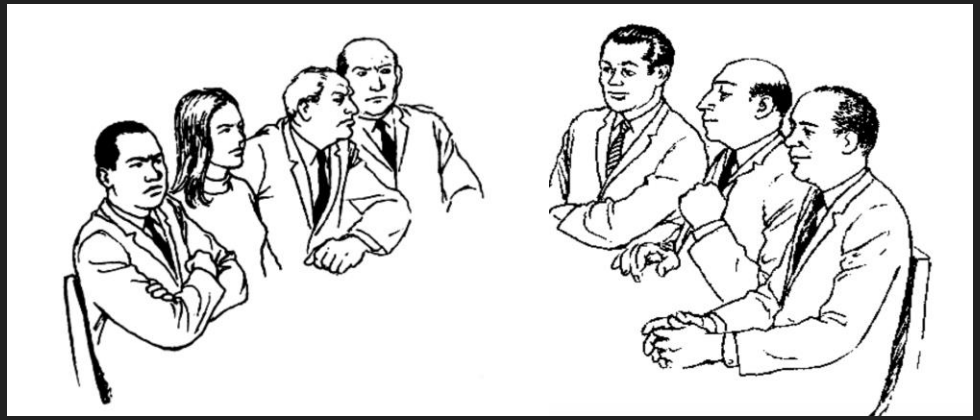


Body language



In this situation, who has the higher position?

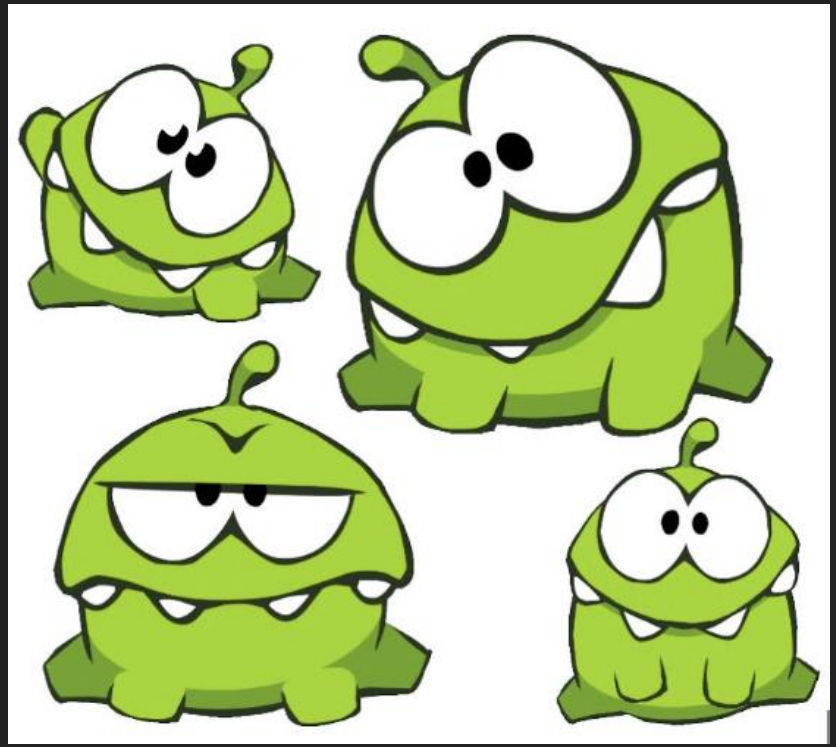
Body language



Wat is going on in this situation?

Virtual pet animations

- Posture: Does your pet have its head held high or low? Is its back straight or hunched?
- Stance: Are its feet close together, indicating shyness, or spread wide, showing confidence?
- Movement: Does it walk with slow, heavy steps,or light, quick ones?



Acting





Animators often use acting as a tool.

Exercise: Body Language in Action

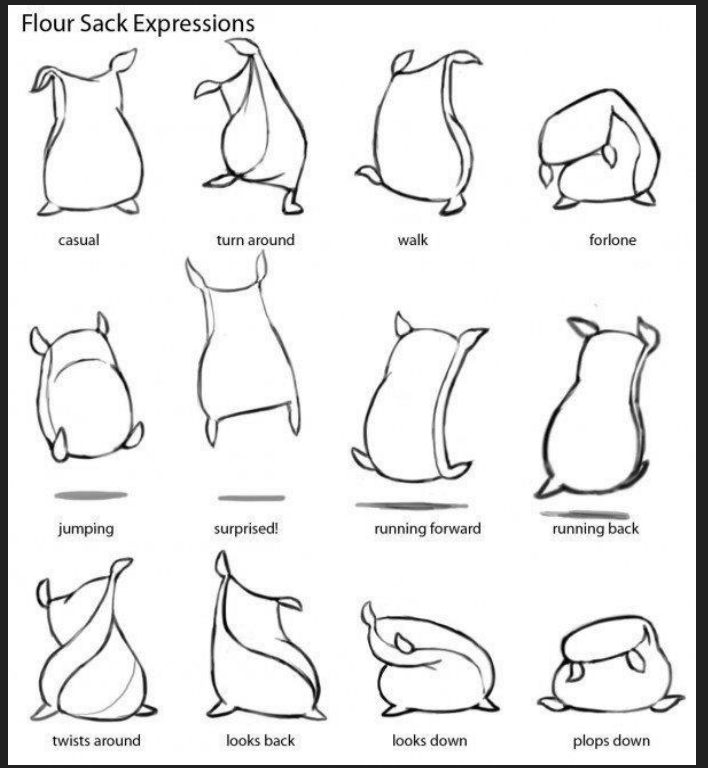
Now that you've seen how animators use their bodies to understand movement, it's your turn to explore. This exercise will help you connect abstract emotions to physical posture.

Work in pairs. Your task is to act out and capture photos of different moods.

- Choose a Mood: chill, depressed, excited, shy, silly, shocked, focused and frustrated.
- Act it out and take a picture
- Analyze: what is the body language communicating? which specific details in your posture make the emotion clear?

The Flour Sack Exercise: A Disney Classic

The "Flour Sack Exercise" is a classic animation exercise that was developed at Disney in the 1930s.



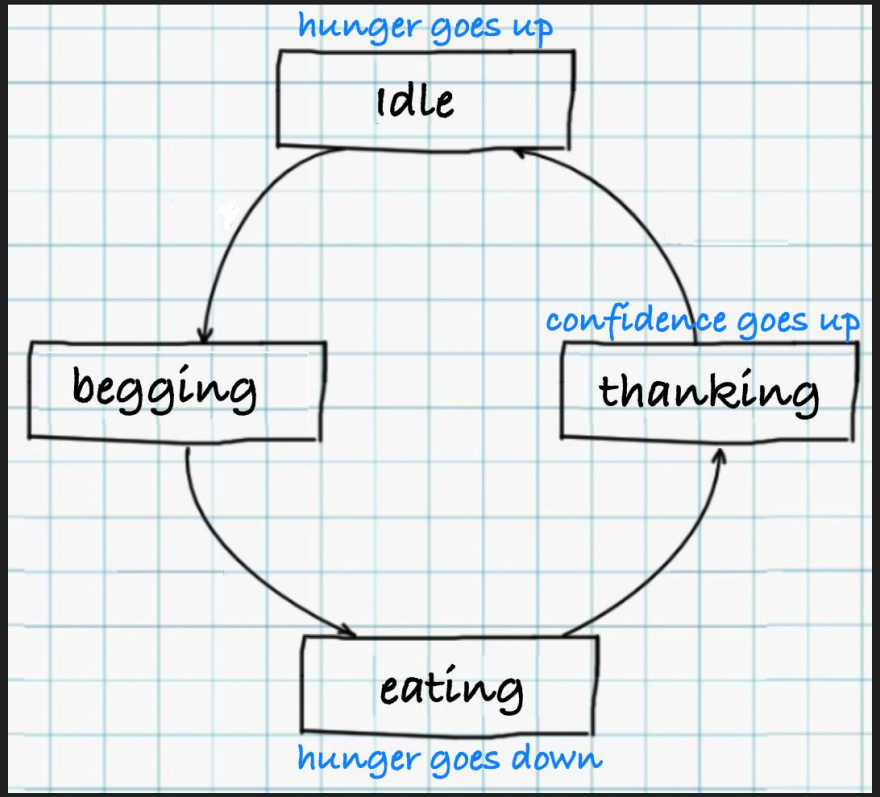
Exercise: Acting with an Inanimate Object

- **Grab an inanimate object.** This could be a blanket, a sweater, or even a pillow.
- **Choose one of the moods from our list:** Shy, Depressed, Silly, or Lonely.
- **Make the object express that emotion and take a picture of your creation.**

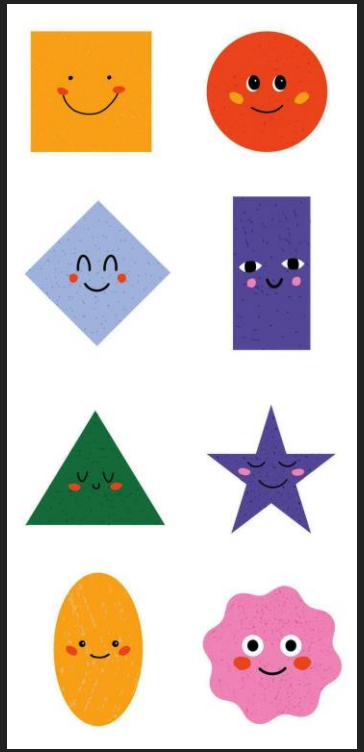


Case example

Activities	Shy Mood	Confident Mood
Idle	Hides or stays in a corner.	Wanders around, plays with toys.
Begging	Cries softly from a distance.	Barks enthusiastically, shoves its tray.
Eating	Eats quickly and nervously.	Eats calmly and enjoys its food.
Thanking	Gives a quick glance at the player.	Barks briefly, wags its tail.



The assignment: your virtual pet



The goal of this assignment is to create a virtual pet with a believable, clear, and attractive personality.

- Refine the Personality: Choose two to three core emotions for your pet and refine the behavioral matrix you created earlier. Think about how the pet behaves when in different moods.
- Design the UI: Create a simple UI in Unity with buttons that correspond to player interactions (e.g., Feed, Play, Cuddle).
- Build the State Diagram: Create a visual representation of your pet's different moods and activities and how it transitions between them.
- Animate the Character: Design a visual concept of the pet, using primitive shapes and colors that suits your pet's personality.

Make sure your pet shows the emotions you designed using as many of the principles of non-verbal communication as possible.

Full description on Brightspace!

Today's Goals: The Virtual Pet Project

1. Playtesting and Documentation

   Test your prototype: Get at least 10 different people to test your virtual pet. Document the results and collect specific feedback about their experience and emotional response to the pet. Document your findings: Summarize the playtest results and describe how you'll use the feedback to improve the player experience.

2. Character and Design

   Refine your FSM: Work with a teammate to discuss and refine each other's Finite State Machine (FSM). Make sure it reflects your design goals and intended personality. Improve animations: Refine the basic animations for each state to create a cohesive design that contributes to a strong sense of personality. Enhance visuals: Ensure your visual design, which should be made from basic geometric shapes, demonstrates a clear understanding of baby schema, color, and shape theory.

3. Submission and Finalization

   Create your demonstration video: Shoot a video (max 3 minutes) that shows the pet's behavior and includes a short explanation of how your state machine works. Write your analysis document: Work on the PDF document, detailing the pet's personality, a state diagram, and how you worked to evoke a specific player emotion. Prepare your build: Make sure your working Unity build is ready to be uploaded to itch.io.

9:15	Iteration 1
11:15	Playtest
12:00	Break
13:00	Iteration 2
14:45	End

Today's Goals: The Virtual Pet Project

Today we can work on the assignment and talk about your progress.
After lunch break, we have a short testing session, so you can go around and test each others pets.